

Answers

7 a $\log_3 5x = \log_3 (2x + 3)$

$$5x = 2x + 3$$

$$x = 1$$

c $\log_4 \frac{x}{x-1} = \log_4 3 + \log_4 2 = \log_4 6$

$$\frac{x}{x-1} = 6$$

$$x = 6x - 6$$

$$x = \frac{6}{5}$$

e $\log_6 x^2 = \log_6 5(2x - 5)$

$$x^2 = 5(2x - 5)$$

$$x^2 - 10x + 25 = 0$$

$$(x - 5)^2 = 0$$

$$x = 5$$

b $\log_9 10x = \frac{3}{2}$

$$10x = 9^{\frac{3}{2}} = 27$$

$$x = 2.7$$

d $\log_5 \frac{5x}{x+2} = \log_5 \frac{x+6}{x}$

$$\frac{5x}{x+2} = \frac{x+6}{x}$$

$$5x^2 = (x+2)(x+6) = x^2 + 8x + 12$$

$$x^2 - 2x - 3 = 0$$

$$(x+1)(x-3) = 0$$

$$x = -1, 3$$

$$\log_5 x \text{ not real for } x = -1 \therefore x = 3$$

f $\log_7 4x - \log_7 \frac{1}{x-6} = 1$

$$\log_7 4x(x-6) = 1$$

$$4x(x-6) = 7$$

$$4x^2 - 24x - 7 = 0$$

$$x = \frac{24 \pm \sqrt{576 + 112}}{8} = 3 \pm \frac{1}{2}\sqrt{43}$$

$$\log_7 4x \text{ not real for } x = 3 - \frac{1}{2}\sqrt{43}$$

$$\therefore x = 3 + \frac{1}{2}\sqrt{43} \quad [= 6.28 \text{ (3sf)}]$$

8 a $\log_x y = 2 \Rightarrow y = x^2$

$$\text{sub. } x^3 = 27$$

$$x = 3$$

$$\therefore x = 3, y = 9$$

b $\log_5 x - 2 \log_5 y = \log_5 2 \Rightarrow \frac{x}{y^2} = 2$

$$\Rightarrow x = 2y^2$$

$$\text{sub. } 3y^2 = 12$$

$$y^2 = 4$$

$$\text{for real } \log_5 y, y > 0 \therefore y = 2$$

$$\therefore x = 8, y = 2$$

c $\log_y 32 = -\frac{5}{2} \Rightarrow y^{-\frac{5}{2}} = 32$

$$\Rightarrow y = 32^{-\frac{2}{5}} = \frac{1}{4}$$

$$\text{sub. } \log_2 x = 3 - 2 \log_2 \frac{1}{4}$$

$$\log_2 x = 3 - (-4) = 7$$

$$x = 2^7 = 128$$

$$\therefore x = 128, y = \frac{1}{4}$$

d $\log_y x = \frac{3}{2} \Rightarrow y^{\frac{3}{2}} = x$

$$\Rightarrow y^{\frac{1}{2}} = x^{\frac{1}{3}}$$

$$\text{sub. } 4x^{\frac{1}{3}} = 20$$

$$x^{\frac{1}{3}} = 5$$

$$x = 5^3 = 125$$

$$\therefore x = 125, y = 25$$

e $\log_a x + \log_a 3 = \frac{1}{2} \log_a y \Rightarrow 3x = y^{\frac{1}{2}}$

$$\Rightarrow y = 9x^2$$

$$\text{sub. } 3x + 9x^2 = 20$$

$$9x^2 + 3x - 20 = 0$$

$$(3x+5)(3x-4) = 0$$

$$\text{for real } \log_a x, x > 0 \therefore x = \frac{4}{3}$$

$$\therefore x = \frac{4}{3}, y = 16$$

f $\log_{10} y + 2 \log_{10} x = 3 \Rightarrow x^2 y = 10^3$

$$\log_2 y - \log_2 x = 3 \Rightarrow \frac{y}{x} = 2^3$$

$$\Rightarrow y = 8x$$

$$\text{sub. } 8x^3 = 1000$$

$$x^3 = 125$$

$$x = 5$$

$$\therefore x = 5, y = 40$$